

Element C: Presentation & Justification of Solution Design Requirements

The major stakeholders are school teachers who will assist students in facilitating engagement and social interaction. The design requirements cater to the needs of the stakeholders by ensuring safety precautions and making sure it is easy for students to use. The stakeholders also requested to ensure the prototype is movable for teachers and students. Given the primary demographics regarding the stakeholders, a prioritized list was developed that reflected the problems associated with student engagement during physical education activities:

- The launcher shall launch the disc 30 feet in terms of horizontal distance. This will allow for the special needs student to interact with other students and make them seem more approachable, as they now have that ability to launch the frisbee a certain distance.
- The launcher should be lightweight and consist of minimal parts to allow for the launcher to be moved around and repaired easily. This feature will allow for teachers, advisors, and even students to help with the repair of the frisbee in the case a malfunction occurs. Battery Dependent: Since the launcher is maneuverable, it requires the use of battery based operation and the launcher must be able to run for 45 minutes. This will allow for the students to enjoy their time interacting with the launcher as well as other students within the period of time assigned for the class period. In addition, the battery should be able to be replaced easily or incorporate the use of a rechargeable feature.
- Sound output (Decibels): The frisbee launcher should be running under 50 Decibels to allow for safe operation for humans, and so that the launcher is not alarming to the students as well as the teachers.
- Cost: The launcher should be made with parts that are cost effective as in the case that a specific part is broken, it can be easily found and replaced so that the feasibility of the launcher is not affected. For example, most of the parts associated with connecting joints, and the body can be manufactured using 3D printing where we can ensure that each part is replicable.
- Appearance: By the frisbee launcher being visually appealing, we can stimulate the visual senses of the special needs students so that they are enticed to use the launcher itself. This allows for the launcher to be used to its full potential and considers all factors associated with pleasing the students